Methylation-driven transcriptional silencing in clear-cell renal cell carcinoma: paired-sample integration of TCGA-KIRC DNA methylation and expression

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Abstract

Background. Paired tumor–adjacent-normal cancer methylation analyses on the Illumina 450K platform routinely declare more than half of tested CpGs differentially methylated at FDR < 0.05, generating lists whose biological actionability is limited. The mechanistically interesting question is not which CpGs are differentially methylated but which methylation changes drive transcriptional silencing of their cognate gene — the epigenetic counterpart of Knudson's tumor-suppressor second hit. **Methods.** We integrate paired DNA methylation and RNA-seq from TCGA-KIRC across 342 (patient, sample_type) keys with both modalities (318 tumor + 24 adjacent-normal), test 137,451 promoter (CpG, gene) pairs by partial-correlation regression of $\log_2(\mathrm{TPM}+1)$ on β adjusting for sample type, age, sex, and tissue-source-site, and define a CpG as silencing when two independent tests agree: a paired tumor-vs-normal moderated test for differential methylation (DM) and a within-status partial-correlation test for methylation–expression association (MEA), both at FDR < 0.05 and with concordant direction (hypermethylation in tumor and high β predicting low mRNA). **Results.** The screen identifies **12,922 silencing (CpG, gene) pairs in 4,483 unique genes (9.4% of tested pairs)** and recovers 13 of 18 predefined ccRCC tumor-suppressor biomarkers as methylation–expression-coupled (10 in the silencing direction; hypergeometric enrichment 1.96×, $p = 0.014$, modest but significant). Most silencing events are subgroup-specific (10,660 / 12,922 pairs, 82.5%) rather than uniform across the cohort. The canonical example is VHL/cg13672843: 32 of 318 tumors (10.1%) exceed the responder threshold, and these tumors show a 33% reduction in VHL mRNA on the linear scale (Wilcoxon $p = 4.5 \times 10^{-6}$, Cohen's $d = -0.79$). Pathway enrichment on the silencing-restricted gene list recovers epithelial–mesenchymal transition, TGF-β signaling, axon guidance, KRAS signaling, and KEGG_PATHWAYS_IN_CANCER (BH-adjusted $p < 0.05$); immune-related pathways that dominate raw-DM GSEA (HALLMARK_INFLAMMATORY_RESPONSE adj. $p = 6 \times 10^{-10}$) drop to non-significance in the silencing-restricted analysis (adj. $p = 1.0$), consistent with the immune signal being a cell-composition artifact that the silencing filter automatically removes. **Conclusion.** Restricting differential methylation to its transcription-coupled subset is a parsimonious way to convert pervasive but uninformative differential-methylation signatures into a functionally interpretable, cell-composition-robust shortlist that recovers canonical cancer-driver loci.

1 Introduction

DNA methylation analyses comparing tumor and adjacent-normal tissue on the Illumina 450K platform routinely yield tens of thousands of FDR-significant cytosine-guanine (CpG) dinucleotide positions. Unlike the sparse-effect regime of germline genome-wide association studies, where typically dozens to hundreds of $\sim 10^6$ tested variants reach genome-wide significance, cancer epigenomes are pervasively disordered: cancer cells exhibit global hypomethylation of repetitive and intergenic DNA combined with focal hypermethylation of CpG islands at tumor-suppressor promoters (Feinberg & Vogelstein 1983; Jones 2012). With $\sim 3 \times 10^5$ CpGs surviving 450K quality control and n on the order of 10^2 paired patients, standard Benjamini-Hochberg correction at $q < 0.05$ declares roughly two-thirds of tested positions significant. The list is statistically valid but biologically uninformative: most differentially methylated CpGs are passenger or compositional changes, not regulators of transcription.

The question that motivates this report is therefore not "which CpGs are differentially methylated" but **"which methylation changes drive transcriptional silencing of their cognate gene"** — the epigenetic counterpart of Knudson's tumor-suppressor second hit and the central paradigm of cancer epigenetics (Jones

& Baylin 2007; Esteller 2008). Hypermethylation of a gene's promoter region is functional only insofar as it suppresses the gene's mRNA output; raw differential methylation is silent on this question.

We address it by integrating paired DNA methylation (450K) and RNA-seq (STAR-Counts, $\log_2(\mathrm{TPM}+1)$) from TCGA-KIRC across 342 (patient, sample-type) keys with both modalities. For each promoter CpG we estimate the within-status association between β and the cognate gene's mRNA and require concordance with the differential-methylation direction before declaring the CpG transcriptionally silencing. Clear-cell renal cell carcinoma (ccRCC) is the natural setting for this analysis: it is the largest paired-design cohort in TCGA, its molecular biology is dominated by VHL inactivation, and its single dominant histology limits biological heterogeneity (TCGA Research Network 2013; Kaelin 2007).

The contribution of this work is to demonstrate that a simple paired methylation–expression integration converts the pervasive 222,386-CpG raw-DM hit list into 12,922 silencing-coupled (CpG, gene) pairs across 4,483 unique genes, recovers VHL/`cg13672843` as a canonical patient-subgroup-driven silencing event, automatically filters out the immune signal that is the dominant — and confounded — top hit of raw-DM GSEA, and identifies a parsimonious shortlist of cancer-relevant pathways enriched among the silenced genes. Methylation–expression integration approaches such as MethylMix (Gevaert 2015) and ELMER (Yao et al. 2015) and the broader expression-quantitative-trait-methylation (eQTM) literature use related correlation screens; the present analysis differs by **(i)** prefiltering on paired-design DM rather than tumor-only DM, **(ii)** requiring the methylation–expression direction to oppose the DM direction (the Knudson-second-hit signature), and **(iii)** treating the two together as a parsimonious cell-composition control that does not require a deconvolution reference panel.

2 Methods

2.1 Cohort, data sources, and preprocessing

DNA methylation 450K β-values, RNA-seq STAR-Counts, and harmonized clinical metadata for TCGA-KIRC were retrieved with `TCGAbiolinks::GDCquery` and `GDCdownload` (Bioconductor 3.22) (Colaprico et al. 2016). The methylation set comprises 484 aliquots from 319 patients (324 primary tumor `-01A` and 160 solid adjacent-normal `-11A`) after exclusion of one `-05A` "additional new primary"; 160 patients have both modalities and form the paired analysis set. The RNA-seq set comprises 614 aliquots, of which 605 unique (patient, sample-type) keys are retained; replicate aliquots are mean-aggregated. Matching the two modalities by (patient, sample-type) yields **342 keys (318 tumor + 24 adjacent-normal)** for the integration analysis. The 485,577-probe β-value matrix was sequentially filtered as in Table 1: cross-reactive probes (Chen et al. 2013) and SNP-overlapping probes (MAF $>$ 0.05 within 2 bp of the 3′ end or the target CpG) were removed with `DMRcate::rmSNPandCH` (Peters et al. 2015), probes with $>$ 20% sample missingness were dropped, and complete-case rows were retained so every patient contributes both aliquots to every paired test. β-values were clipped to $[0.001, 0.999]$ and transformed to $M = \log_2(\beta/(1-\beta))$ for variance stabilization (Du et al. 2010).

Table 1: Preprocessing attrition.

Step	Probes retained
Baseline (450K array)	485,577
Intersect with Illumina 450K annotation	472,651
Drop chr X / chr Y	461,295
Cross-reactive + SNP overlap (DMRcate)	433,595
Drop probes with >20% sample missingness	395,325
Complete-case across 484 samples	320,953

2.2 Differential methylation (DM) test

Silencing requires concordant evidence from two independent tests. The **DM test** (this section) asks whether mean methylation differs between paired tumor and adjacent-normal samples and gives the direction (hyper- vs hypomethylation). The **MEA test** (next subsection) asks whether, within sample type, methylation predicts mRNA — the gene-regulatory effect. Naive cross-sample correlation is confounded by the bulk tumor/normal mean shift (*cf.* VHL: $\rho = -0.03$ across all samples vs. within-status partial $p = 2 \times 10^{-6}$); their concordant intersection is the Knudson-second-hit signature.

For each CpG, within-pair M-value differences $D_i = M_i^{\text{tumor}} - M_i^{\text{normal}}$ ($i = 1, \ldots, 160$) were tested against $H_0 : \mathbb{E}[D] = 0$ using `limma::lmFit` with an intercept-only design followed by empirical-Bayes shrinkage via `eBayes` (Smyth 2004; Ritchie et al. 2015). For a balanced paired design this is statistically equivalent to the moderated paired t-test; patient-level covariates cancel in D_i. Benjamini–Hochberg FDR at $q < 0.05$ is used throughout the report unless otherwise stated.

2.3 Methylation–expression association (MEA) test

A promoter (CpG, gene) pair is built from the 450K UCSC RefGene annotation: for each CpG with a non-empty `UCSC_RefGene_Name` we expand to one row per (gene, group) entry, retain pairs where the group is `TSS1500`, `TSS200`, `5'UTR`, or `1stExon`, and de-duplicate so each (CpG, gene) pair appears once with the most-promoter-proximal group label. Pairs are restricted to genes measured in RNA-seq, yielding 137,451 (CpG, gene) pairs across 128,023 unique CpGs and 15,842 unique genes.

For each pair we fit

$$\log_2(\text{TPM}_{\text{gene}} + 1) \;=\; \alpha + \beta\, \beta_{\text{CpG}} + \gamma^{\top} Z + \varepsilon, \qquad Z = (\text{sample\_type, age, sex, TSS})$$

across the 342 matched samples. The `sample_type` covariate absorbs the tumor-vs-normal mean shift in both methylation and expression; the test on β is therefore a within-status partial association. We compute the test analytically as a partial correlation after residualizing $\log_2$TPM and β against the covariate design (QR decomposition on a 15-column design matrix; $df_{\text{resid}} = 326$); this vectorizes to 137,451 pair-tests in <1 second. The per-CpG **responder fraction** is $\hat{\pi}_{\text{resp}} = \frac{1}{n_{\text{tumor}}} \sum_{i \in \text{tumor}} \mathbb{1}\{|\beta_i - \bar{\beta}_{\text{normal}}| > 0.2\}$, the proportion of tumors whose β deviates from the adjacent-normal mean by more than 0.2 (a $\Delta\beta$ corresponding to a clear methylation transition). The threshold 0.2 is the standard biological significance cutoff in 450K analyses; the threshold for the responder fraction itself is discussed below. Conceptually, $\hat{\pi}_{\text{resp}}$ corresponds to a **dominant-encoded** genetic-model treatment of methylation status — methylated above 0.2 / not-methylated — analogous to how an additive-vs-dominant variant encoding determines the design matrix in single-locus association tests.

2.4 Definition of silencing and pattern partition

A (CpG, gene) pair is **silencing** if it satisfies all three: (i) DM FDR < 0.05, (ii) MEA FDR < 0.05, and (iii) $\widehat{\text{logFC}}_{\text{DM}} > 0$ and $\hat{\beta}_{\text{MEA}} < 0$ — that is, hypermethylation in tumor and high β predicting low mRNA. This is the strict Knudson-second-hit direction. The reverse direction (hypomethylation in tumor with high β predicting high mRNA) we label **activation** and report descriptively as a secondary observation; activation pairs are not the focus of this report.

Among silencing CpGs we further partition by responder fraction with a single descriptive cut at 0.30: **subgroup-driven silencing** ($\hat{\pi}_{\text{resp}} < 0.3$, the rare-but-meaningful pattern exemplified by VHL) versus **cohort-wide silencing** ($\hat{\pi}_{\text{resp}} \geq 0.3$, the SFRP1/RASSF1A-style uniform pattern). The 0.30 cut is operational, not a biological law; the responder fraction is best read as a continuous descriptor.

2.5 Pathway enrichment

Two pathway analyses are performed against MSigDB Hallmark (50 sets) and KEGG Legacy including `KEGG_RENAL_CELL_CARCINOMA` (Subramanian et al. 2005; Liberzon et al. 2015). (i) **GSEA**: each gene is ranked by $\text{sign}(\hat{\beta}) \cdot -\log_{10}(p_{\text{assoc}})$ at its best (lowest-p) promoter CpG, so that strongly silencing genes have strongly negative ranks; we use `fgsea` (Korotkevich et al. 2021). Selecting one lead CpG per gene is the omics analog of LD clumping in GWAS: correlated promoter CpGs within a gene share local DNA-methylation structure, and collapsing to a single representative removes redundant signal in the gene-level ranking. (ii) **ORA**: the silencing gene set is the foreground, all 15,842 tested genes are the background, and pathway enrichment is tested by hypergeometric distribution. The raw-DM GSEA serves as the comparator to evaluate whether immune-pathway signals dominant in raw DM disappear under the silencing restriction.

2.6 Validation, calibration, and predefined controls

Literature biomarker recovery (sanity check): an 18-gene biomarker list curated from the ccRCC methylation literature provides a benchmark; the silencing screen should recover canonical TSGs such as VHL, RASSF1A, SFRP1/2, GATA5, and DKK3.

VHL positive control: at least one VHL-promoter CpG must be flagged as silencing.

Calibration: the genomic inflation λ_{GC} for the DM test is 36.8, well above the GWAS rule-of-thumb of 1.10 (Devlin & Roeder 1999). A sign-flipping permutation null on 30,000 probes $\times$ 20 replicates gives $\lambda_{GC} = 1.0 \pm 0.4$, indicating that the inflation reflects pervasive real signal rather than miscalibration.

3 Results

3.1 Cohort structure (Figure 1)

To characterize cohort structure and identify systematic technical or biological variation that could confound DM, we performed PCA on the top-10,000-variance CpGs (Figure 1). PC1 (32.5% variance) cleanly separates tumor from adjacent-normal; PC2 (19.6%) tracks sex (one-way ANOVA $R^2 = 0.94$), motivating chr X / chr Y exclusion in preprocessing. Lower PCs reflect tumor grade, tissue-source-site, and plate batch.

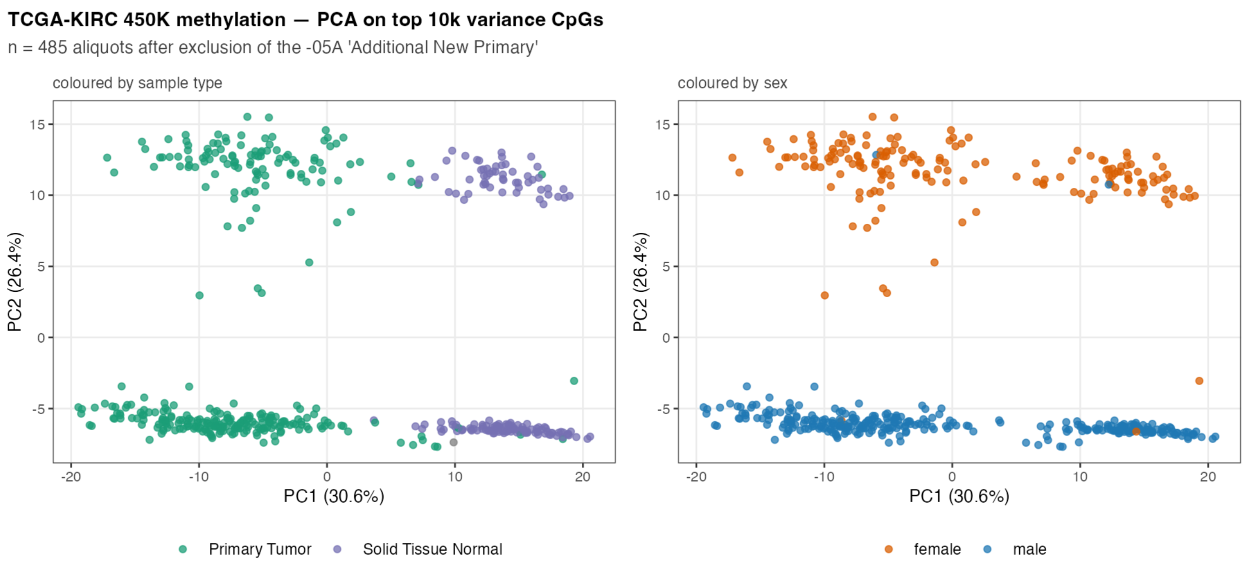

Figure 1: PCA on the top 10,000 most-variable CpGs (pre-QC EDA panel; $n = 484$ samples after exclusion of the single "-05A" aliquot). Left: tumor vs adjacent-normal separation along PC1. Right: same scatter coloured by recorded sex, showing PC2 tracks the sex axis ($R^2 = 0.94$).

3.2 DM test: paired differential methylation is pervasive but uninformative on its own (Figure 2)

Moderated paired tests on 320,953 CpGs identify **222,386 FDR-significant positions (69.3% of tested)**, near-balanced between hypermethylated (107,277) and hypomethylated (115,109). The p-value histogram (Figure 2, top-left) shows a massive density spike near 0 and elevation across all bins, while the sign-flip permutation null is flat at density ≈ 1 — directly visualizing that the inflation ($\lambda_{GC}^{\text{real}} = 36.8$ vs $\lambda_{GC}^{\text{perm}} = 0.95$) is real pervasive signal, not test miscalibration. The paired-$\Delta\beta$ distribution (top-right) shows most CpGs at $|\Delta\beta| < 0.1$ with 9,312 (2.9%) clearing the 0.2 biological cutoff, so most of the 222,386 FDR hits are biologically small. A list of 222,386 CpGs is therefore not interpretable at the gene-regulatory level on its own, motivating the integration filter below.

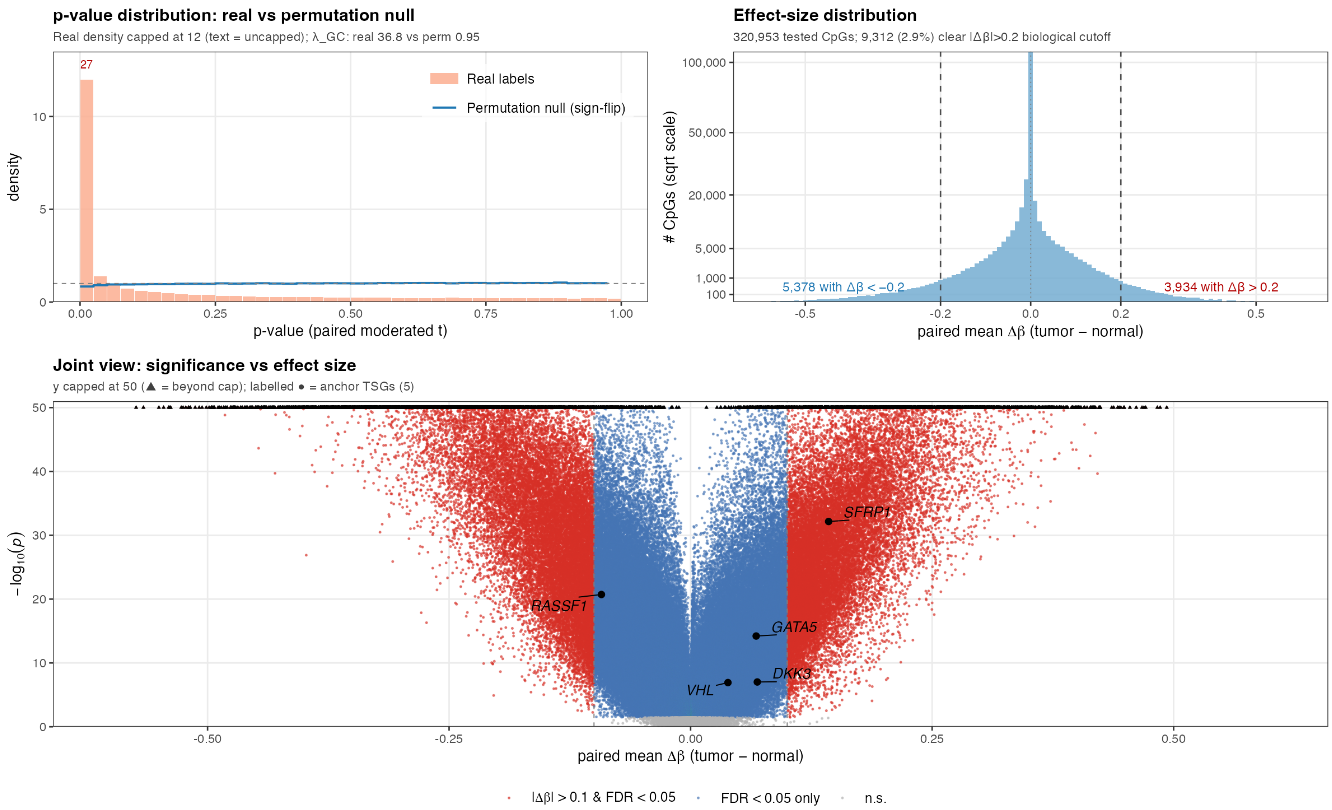

Figure 2: Paired tumor–normal DM diagnostic. Top-left: p-value distribution — real labels (peach bars; density capped at 12 for plotting, the leftmost bin reaches 27) show massive concentration near 0 with elevation across all bins, while the sign-flip permutation null (blue line) is flat at density ≈ 1 ($\lambda_{GC}^{\text{real}} = 36.8$ vs $\lambda_{GC}^{\text{perm}} = 0.95$); the contrast is direct visual evidence that the test is calibrated and the inflation is real signal. Top-right: paired-mean $\Delta\beta$ distribution on a sqrt scale; most mass sits in $|\Delta\beta| < 0.1$, with 9,312 (2.9%) clearing $|\Delta\beta| > 0.2$ (the standard 450K biological-significance cutoff). Bottom: volcano on paired mean $\Delta\beta$ (y capped at 50; triangles = beyond cap); five anchor TSGs labelled — SFRP1/GATA5/DKK3 in the cohort-wide hypermethylation cluster, VHL near the origin (subgroup-driven; cohort-mean $\Delta\beta \approx 0$), RASSF1 on the hypomethylation side (§4.2 anomaly).

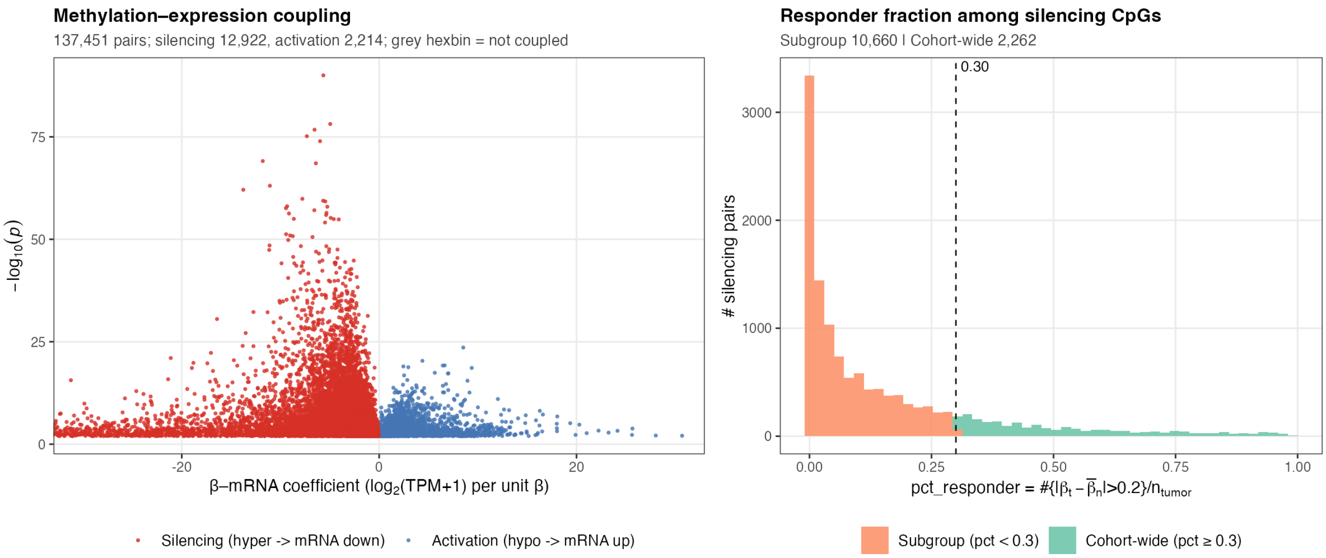

Figure 3: Methylation–expression coupling at promoter CpGs. Left: β–mRNA volcano; red = silencing (hypermethylation in tumor and within-cohort $\beta\uparrow \Rightarrow$ mRNA $\downarrow$); blue = activation (the reverse direction); grey = not coupled. Right: distribution of responder fraction among silencing pairs; most silencing (82.5%) is patient-subgroup-driven (orange, $\hat{\pi}_{\text{resp}} < 0.3$) rather than cohort-wide (green, ≥ 0.3).

3.3 Methylation–expression integration identifies 12,922 silencing (CpG, gene) pairs (Figure 3)

Across **137,451 promoter (CpG, gene) pairs**, 80,179 (58.3%) reach DM FDR < 0.05 and 30,651 (22.3%) reach MEA FDR < 0.05. Imposing the strict-silencing direction yields **12,922 silencing pairs (9.4% of tested)** spanning 12,701 unique CpGs and 4,483 unique genes (Figure 3, left). The reverse-direction *activation* set (hypomethylation $\cap$ positive β–mRNA coefficient) contains 2,214 pairs (silencing : activation $\approx 6:1$, consistent with hypermethylation-driven silencing being the dominant epigenetic cancer mechanism) and is reported descriptively; activation pairs are not the focus of this report.

3.4 Most cancer methylation silencing is patient-subgroup-specific (Figure 3 right; Figure 4)

We next ask whether the 12,922 silencing pairs represent a uniform pattern across the cohort or a heterogeneous landscape in which different patient subsets silence different loci. The responder-fraction distribution is monotone right-skewed (Figure 3, right): **10,660 of 12,922 silencing pairs (82.5%) have $\hat{\pi}_{\text{resp}}$** < 0.3, meaning they are silenced in only a minority of tumors; the remaining 17.5% are cohort-wide. This skew is the central descriptive finding: most silencing CpGs affect patient subgroups rather than the entire cohort, and mean-based DM testing dilutes such signals. Figure 4 illustrates both patterns at six representative loci. VHL (top-left) shows the canonical subgroup pattern: most tumors cluster at low β but a tail of $\sim 10\%$ are strongly hypermethylated and have low VHL mRNA. SFRP1 (top-middle) shows the cohort-wide pattern: the majority of tumors are hypermethylated and have low SFRP1 mRNA, while normals cluster at low β and high mRNA. DKK3, GATA5, and PCDH17 illustrate intermediate patterns. Note that for VHL the *cross-sample* Spearman ρ is -0.03 — the linear correlation across all 342 samples is misleading because it is driven by the bulk of tumors near zero — yet the within-status partial association p-value is 2×10^{-6}. This is the analytical justification for using a regression with `sample_type` as a covariate rather than naive correlation.

3.5 Pathway enrichment: the silencing filter as a reference-panel-free cell-composition control (Figure 5)

Bulk tumor DM is confounded by cell composition: tumors typically carry more infiltrating leukocytes than adjacent normal kidney, so any CpG that distinguishes immune from epithelial cells appears differentially methylated for compositional reasons alone, with no real epigenetic change in the tumor cells themselves. The standard fix — reference-based deconvolution (EpiDISH, CIBERSORT) — adds a tool dependency and

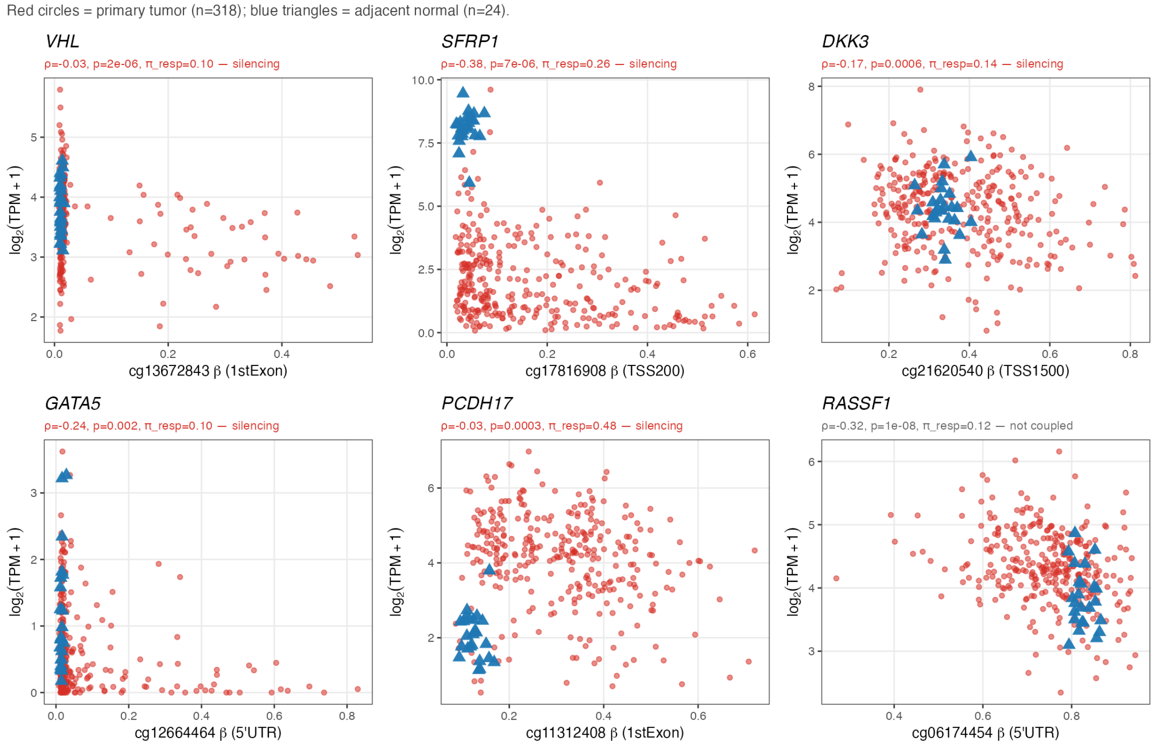

Figure 4: Promoter β vs gene expression at six representative loci. Each panel uses the per-gene best-association silencing CpG (or, for RASSF1, best-association non-silencing CpG). VHL/cg13672843 = subgroup pattern; SFRP1, DKK3, GATA5, PCDH17 = silencing across patient subsets; RASSF1 = ccRCC-specific anomaly (see §4.2).

requires a tissue-matched reference panel. We hypothesize that silencing's directional concordance criterion automatically excises this signal: an immune-cell CpG bulk-hypomethylated in tumor *by* leukocyte infiltration also carries elevated bulk mRNA from the same infiltrating cells — the *opposite* of silencing's hypermethylation + low-mRNA signature — so it fails directional concordance and never enters the silencing list. **Prediction**: pathways enriched in raw-DM purely on cell-composition grounds (notably immune pathways) should drop out of silencing-restricted enrichment.

We tested this by comparing GSEA on the raw-DM ranking against ORA on the silencing gene list. The raw-DM top hits form an eight-pathway immune block — HALLMARK_INFLAMMATORY_RESPONSE ($p_{adj} = 6.0 \times 10^{-10}$, NES -2.21), Complement (3.4×10^{-6}), Allograft Rejection (9.6×10^{-6}), Interferon γ Response (1.1×10^{-5}), TNFα NF-κB (4.0×10^{-5}), Interferon α Response (1.4×10^{-3}), KEGG Hematopoietic Cell Lineage (2.9×10^{-3}), IL6 JAK STAT3 ($p_{adj} = 0.25$, included for completeness). In the silencing-restricted ORA, **all eight collapse to** $p_{adj} = 1.0$ — HALLMARK_INFLAMMATORY_RESPONSE alone moves ten orders of magnitude (Table 2; Figure 5). The prediction is confirmed at striking magnitude.

What survives the silencing filter is recognizable cancer biology: Hallmark EMT ($p_{adj} = 3.2 \times 10^{-7}$, the canonical invasion–metastasis program), KEGG Axon Guidance (1.4×10^{-4}, dominated by the protocadherin cluster — a known ccRCC silencing target), KEGG TGF-β Signaling (8.1×10^{-4}), Hallmark KRAS Signaling Up (7.9×10^{-3}), KEGG Pathways in Cancer (7.6×10^{-3}), KEGG ECM Receptor Interaction (4.7×10^{-3}). As a bonus check, KEGG_OLFACTORY_TRANSDUCTION — KEGG's largest gene set (~400 olfactory receptors clustered on chromosome 11p15, sharing low expression and a notorious large-family GSEA artifact) — also tops the raw-DM list and disappears under the silencing restriction. The silencing filter therefore clears at least two distinct GSEA failure modes (cell-composition confounding and set-size artifact) without an external reference panel. **The methodological claim of this report is concrete: a single per-CpG regression of $\log_2$(TPM) on β adjusted for sample type, combined with directional concordance, substitutes for an external deconvolution reference panel.**

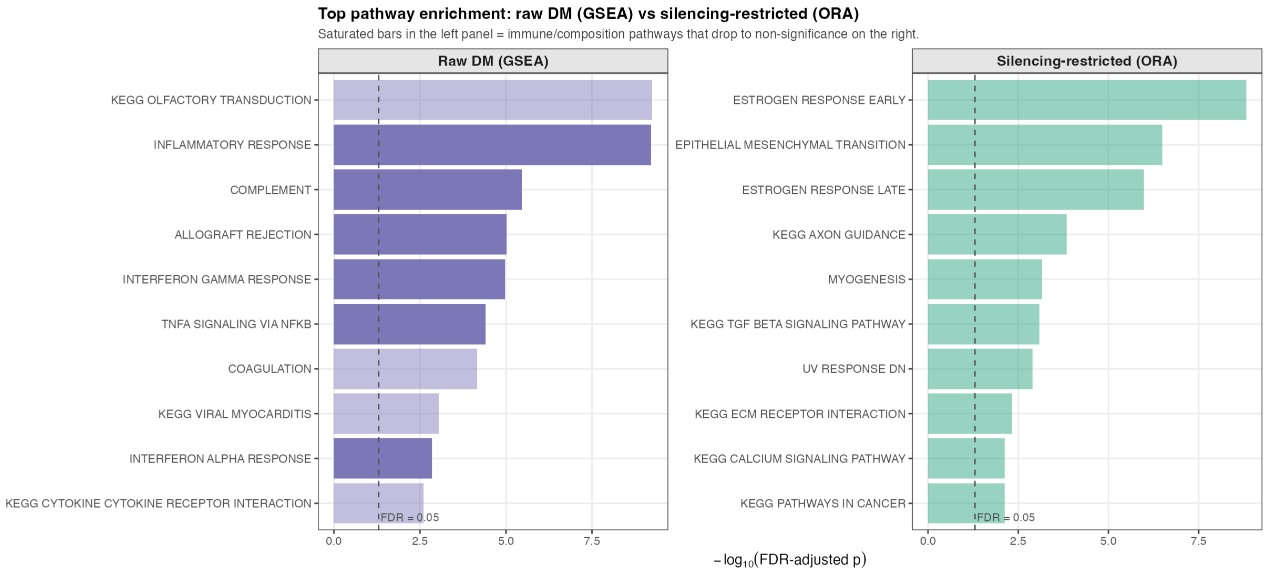

Figure 5: Top-10 pathway enrichment hits, raw-DM GSEA (left) vs silencing-restricted ORA (right). Raw-DM top hits are dominated by immune and coagulation sets (cell-composition-driven) plus KEGG_OLFACTORY_TRANSDUCTION (a known large-family GSEA artifact); silencing-restricted top hits are cancer-relevant (EMT, TGF-β, axon guidance, KRAS, KEGG Pathways in Cancer). The two panels use different tests (signed GSEA on the left, unsigned hypergeometric ORA on the right) because GSEA requires a per-gene ranking that is undefined for the silencing foreground; the comparison is intended to contrast top-hit *content*, not test statistics.

Table 2: Immune-related pathways: collapse of significance from raw-DM GSEA to silencing-restricted ORA. k = silencing foreground genes in the set; K = total set size in the tested-gene background.

Pathway	Raw-DM GSEA padj	Raw-DM NES	Silencing ORA padj	k / K
HALLMARK INFLAMMATORY RESPONSE	6.0e-10	-2.21	1	33 / 185
HALLMARK COMPLEMENT	3.4e-06	-1.98	1	50 / 179
HALLMARK ALLOGRAFT REJECTION	9.6e-06	-1.98	1	28 / 173
HALLMARK INTERFERON GAMMA RESPONSE	1.1e-05	-1.93	1	32 / 182
HALLMARK TNFA SIGNALING VIA NFKB	4.0e-05	-1.85	1	38 / 189
HALLMARK INTERFERON ALPHA RESPONSE	1.4e-03	-1.83	1	13 / 88
KEGG HEMATOPOIETIC CELL LINEAGE	2.9e-03	-1.85	1	21 / 75
HALLMARK IL6 JAK STAT3 SIGNALING	2.5e-01	-1.40	1	16 / 78

3.6 VHL/cg13672843: biological exemplar of subgroup-driven silencing (Figure 6)

VHL is the canonical exemplar of why mean-based DM and within-status integration disagree. Its 1stExon CpG `cg13672843` is the predefined positive control of the analysis: across all 318 tumors the cohort-mean $\Delta\beta$ is only 0.04 (essentially invisible in the volcano of Figure 2), yet 32 (10.1%) tumors carry $\beta \geq 0.2$ and these responders show **a 33% reduction in VHL mRNA on the linear scale** (mean tumor TPM 8.7 vs 12.9; Wilcoxon $p = 4.5 \times 10^{-6}$, Cohen's $d = -0.79$; Figure 6). The within-status partial-association $p = 2 \times 10^{-6}$ contrasts sharply with the cross-cohort Spearman $\rho = -0.03$, confirming the analytical justification for adjusting `sample_type` in the MEA test (Methods §2.3). The 32-tumor responder fraction matches the documented ~15% methylation-silencing rate for VHL in ccRCC (Herman et al. 1994; Kaelin 2007); the 24 patients with paired aliquots in both modalities show the same direction of effect ($\rho = -0.36$ between $\Delta\beta$ and $\Delta \log_2$VHL TPM, $p = 0.09$, underpowered).

3.7 Threshold sensitivity (Table 3)

We test whether the silencing definition is driven by the choice of FDR cutoff. A 3×3 grid over DM and MEA FDR cutoffs $\in \{0.01, 0.05, 0.10\}$ (Table 3) shows the silencing pair count varying from 8,914 to 15,751 (1.8×), but **the top-5 ORA pathway hits are identical to within rank order across all 9 cells**: EMT, ESTROGEN_RESPONSE_EARLY, ESTROGEN_RESPONSE_LATE, AXON_GUIDANCE, MYOGENESIS appear in every combination. The pattern partition is similarly stable: at $\hat{\pi}_{\text{resp}}$ cuts of 0.20 /

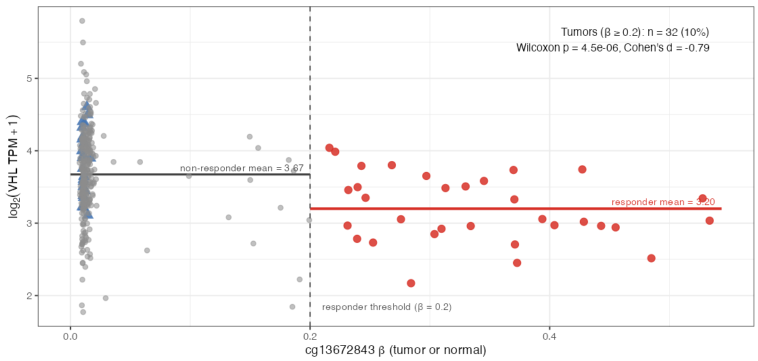

Figure 6: VHL/cg13672843 as a subgroup-driven silencing event. Red = responder tumors ($\beta \geq 0.2$, $n = 32$); grey = non-responder tumors ($n = 286$); blue = adjacent normals ($n = 24$). Responders show 33% lower VHL mRNA on the linear scale (Wilcoxon $p = 4.5 \times 10^{-6}$, Cohen's $d = -0.79$).

0.30 / 0.40 the subgroup share is 73% / 82% / 89%, leaving the qualitative conclusion (silencing is dominantly subgroup-driven) unchanged across reasonable thresholds.

3.8 Literature biomarker recovery (Table 4)

We next ask whether the silencing screen recovers known ccRCC tumor-suppressor genes silenced by methylation. Of 18 biomarkers compiled from the literature — a heterogeneous heuristic rather than a curated gold standard, with entries ranging from extensively documented (VHL, RASSF1) to single-cohort weak reports — 13 (72%) have at least one strict-silencing or activation-direction promoter CpG and 10 (56%) have at least one strict-silencing CpG (Table 4). Because the silencing list spans 4,483 / 15,842 tested genes (28.3%), the random null expects only 5.1 / 18 biomarkers recovered, so the observed counts correspond to **hypergeometric enrichment of** $2.55\times$ **(**$p = 1.4 \times 10^{-4}$**) for any coupling and** $1.96\times$ **(**$p = 0.014$**) for silencing only**. The enrichment is statistically significant but modest in magnitude, and we read it cautiously: the 18-gene benchmark mixes well-evidenced and weak entries (Discussion §4.2). VHL, SFRP1/2, GATA5, DKK3, PCDH17, COL14A1, SLIT2, TIMP3, and UCHL1 are recovered as silencing; five genes (PITX2, APAF1, SFRP4, CDKN2A, NEFH) have neither silencing nor activation; RASSF1 is the conspicuous direction anomaly discussed in §4.2.

3.9 External replication in TCGA-KIRP (papillary RCC)

We tested whether the silencing pipeline replicates in an independent RCC subtype. Applying the identical pipeline (paired-design DM + within-status MEA + directional concordance, FDR thresholds 0.05/0.05) to 45 paired KIRP patients yields 9,116 silencing pairs across 2,955 unique genes — 66% of the KIRC silencing-gene count, reflecting the smaller paired n. The 1,889-gene cross-cohort overlap is highly enriched over random sampling (Jaccard 0.34, hypergeometric $p < 10^{-300}$), supporting reproducibility of the silencing screen.

The biomarker triangulation is informative for interpretation. Of the 18 literature ccRCC TSGs, 5 are shared between cohorts (SFRP1, DKK3, SLIT2, TIMP3, UCHL1 — robust pan-RCC TSGs), 5 recover in KIRC only — most importantly **VHL**, consistent with VHL being a ccRCC-specific driver — and 2 in KIRP only. Notably, **RASSF1 recovers in KIRP but not KIRC**, providing independent external evidence that the KIRC RASSF1 reversal (§4.2) is a genuine ccRCC-specific anomaly rather than a methodological failure. The 6 genes that fail to recover as silencing in **either** cohort (PITX2, APAF1, SFRP4, CDKN2A, NEFH, SFRP5) are most parsimoniously explained by benchmark-list heterogeneity (CDKN2A is dominated by deletion/mutation in ccRCC; the others rely on single-cohort or weak literature support).

The apparent KIRP–KIRC silencing-list size gap is driven by statistical power, not methodological sensitivity. Under an effect-size-anchored threshold scheme (DM FDR < 0.10, MEA FDR < 0.10, $|\widehat{\log\mathrm{FC}}_{\mathrm{DM}}| > 0.5$ on the M scale), KIRP and KIRC yield 2,879 and 2,779 silencing genes respectively (KIRP/KIRC ratio 104%), confirming that the substantive output is effect-size driven and effectively cohort-size independent once the FDR-power confound is removed.

Table 3: Sensitivity of silencing detection and top-5 ORA hits to DM/MEA FDR cutoffs (default = 0.05/0.05). Pathway names abbreviated: EMT = EPITHELIAL_MESENCHYMAL_TRANSITION; ESTR. = ESTROGEN_RESPONSE_; ECM_RECEPTOR = ECM_RECEPTOR_INTERACTION.

DM FDR	MEA FDR	# silencing pairs	# unique genes	Top 5 ORA hits (in order)
0.01	0.01	8,914	3,266	EMT / ESTR.EARLY / ESTR.LATE / AXON_GUIDANCE / MYOGENESIS
0.01	0.05	11,829	4,144	EMT / ESTR.EARLY / ESTR.LATE / AXON_GUIDANCE / ECM_RECEPTOR
0.01	0.10	13,746	4,692	EMT / ESTR.EARLY / ESTR.LATE / AXON_GUIDANCE / MYOGENESIS
0.05	0.01	9,617	3,483	ESTR.EARLY / ESTR.LATE / EMT / MYOGENESIS / AXON_GUIDANCE
0.05	0.05	12,922	4,483	ESTR.EARLY / EMT / ESTR.LATE / AXON_GUIDANCE / MYOGENESIS
0.05	0.10	15,104	5,115	ESTR.EARLY / EMT / ESTR.LATE / AXON_GUIDANCE / MYOGENESIS
0.10	0.01	9,938	3,587	ESTR.EARLY / ESTR.LATE / EMT / UV_RESPONSE_DN / MYOGENESIS
0.10	0.05	13,418	4,642	ESTR.EARLY / EMT / ESTR.LATE / AXON_GUIDANCE / UV_RESPONSE_DN
0.10	0.10	15,751	5,317	EMT / ESTR.EARLY / ESTR.LATE / MYOGENESIS / AXON_GUIDANCE

Table 4: Silencing recovery for 18 predefined ccRCC methylation TSG biomarkers (# CpG = promoter CpGs tested; # sil / # act = silencing-direction / activation-direction CpGs passing the integration filter).

Gene	# CpG	# sil	# act	Status
RASSF1	37	0	1	Activation only
BNC1	18	0	1	Activation only
SFRP5	16	0	1	Activation only
PITX2	25	0	0	Not coupled
APAF1	21	0	0	Not coupled
SFRP4	11	0	0	Not coupled
CDKN2A	6	0	0	Not coupled
NEFH	6	0	0	Not coupled
TIMP3	13	2	1	Silencing recovered
UCHL1	3	3	0	Silencing recovered
SFRP2	34	4	0	Silencing recovered
VHL	7	1	0	Silencing recovered
SFRP1	15	5	0	Silencing recovered
SLIT2	6	3	0	Silencing recovered
COL14A1	11	2	0	Silencing recovered
PCDH17	13	4	0	Silencing recovered
DKK3	17	5	0	Silencing recovered
GATA5	19	6	0	Silencing recovered

4 Discussion

4.1 Differential methylation versus functional silencing

In paired tumor-vs-normal 450K designs, single-sample standard deviations on the M-scale are small enough that mean $\Delta\beta$ on the order of 0.01 is detectable; two-thirds FDR-significant positions are a biological expectation, not a pipeline failure (the permutation-null $\lambda_{GC} = 1.0$ rules out miscalibration). The practical consequence is that raw FDR-filtering cannot serve as the final filter for biological importance, and some criterion of functional consequence is required. Transcriptional coupling is the natural such criterion since promoter methylation's claim to gene-regulatory significance is precisely its ability to suppress transcription (Jones & Baylin 2007). The filter — one per-CpG regression and one direction check — reduces 222,386 raw-DM hits to 12,922 silencing pairs whose biology is interpretable, and it does so without requiring a deconvolution reference panel. This is the report's central methodological claim.

4.2 Subgroup-driven silencing and the cell-composition fingerprint

The responder-fraction distribution among silencing CpGs is right-skewed with 82.5% of pairs subgroup-driven (Figure 3, right). This contradicts the conventional "common methylation silencing" framing — instead, the typical silencing CpG affects a minority of tumors. The interpretation is consistent with the **alternative-routes-to-inactivation** logic of cancer-suppressor loss: VHL is inactivated in 60–80% of ccRCC tumors but only ~15% by methylation (Kaelin 2007). The present analysis suggests this pattern is the rule, not the exception. Mean-based tests dilute such signals across cohorts; responder-fraction screens are a necessary complement.

The cell-composition control falls out automatically: an immune-marker CpG hypomethylated in tumor *only* *because* tumor tissue contains more infiltrating leukocytes will not show concordant epithelial mRNA change in the bulk RNA-seq of the same sample. The empirical result (Table 2; Figure 5) — Hallmark Inflammatory Response from $p_{adj} = 6\times10^{-10}$ in raw-DM GSEA to 1.0 in silencing-restricted ORA — confirms this directly. The silencing-restricted top hits (EMT, TGF-β, axon guidance, KRAS, KEGG Pathways in Cancer) are recognizable cancer biology, not infiltration biology.

A conspicuous negative finding deserves noting: across RASSF1's 37 promoter CpGs, none passes the silencing filter; one passes activation-direction. RASSF1 is hypomethylated rather than hypermethylated in TCGA-KIRC tumors at every promoter CpG with a strong β–mRNA coupling (Figure 4, bottom-right). The DM best-CpG $\Delta\beta = -0.33$ confirms cohort-level hypomethylation. The canonical RASSF1A methylation-silencing portrayal may be a feature of non-RCC cancers or papillary RCC but does not hold in ccRCC at TCGA scale; more broadly, the 5/18 non-recovered biomarkers in Table 4 include several genes whose ccRCC-specific methylation evidence is from single small cohorts (NEFH, APAF1) and should be read as biomarker-list heterogeneity rather than wholesale method failure.

4.3 Limitations

Bulk-tissue mixing is reduced but not eliminated by the silencing filter; quantitative deconvolution (e.g., EpiDISH) was beyond course scope. *Single platform / single data source*: the KIRP external replication (§3.9) addresses single-cohort concerns within RCC, but both KIRC and KIRP are TCGA Illumina 450K — true platform-independent validation (e.g., CPTAC ccRCC, EPIC array) is not done in this report. *Association is not causation*: methylation may be downstream of expression or share a common cause; directional concordance with paired-design DM and the VHL mechanistic match are consistent with but do not prove causal silencing. *Promoter restriction*: gene-body and enhancer CpGs are excluded; some silencing may operate through non-promoter elements. *Limited paired RNA-seq subset*: only 24 patients have paired tumor and adjacent-normal aliquots in both modalities, so the within-pair $\Delta\beta \times \Delta \log_2 \text{TPM}$ check ($p = 0.09$ for VHL) is power-limited; the primary MEA analysis relies on a within-status partial association across 342 cross-sectional aliquots, not a within-pair difference. *Threshold dependence*: the $\hat{\pi}_{\text{resp}}$ cut at 0.30 and $\Delta\beta$ cut at 0.20 are operational; KIRP replication uses identical pipeline parameters, and the smaller silencing-list size in KIRP reflects FDR-based power scaling (the effect-size-anchored sensitivity check in §3.9 confirms cohort-size-independence of the underlying signal once the FDR-power confound is removed).

5 Conclusion

The central methodological contribution of this report is that silencing — defined as the directional intersection of paired-design DM and within-status MEA — is sufficient to identify transcriptionally silencing CpGs without a cell-composition deconvolution panel. The construct operationalizes Knudson's tumor-suppressor "second hit" in the methylation domain, and its empirical signature is striking: the immune-pathway block dominating raw-DM enrichment (HALLMARK_INFLAMMATORY_RESPONSE $p_{adj} = 6 \times 10^{-10}$) collapses to non-significance under the silencing restriction, with a single per-CpG regression adjusted for sample type substituting for reference-based tools such as EpiDISH or CIBERSORT in this specific confound. On TCGA-KIRC the screen reduces a pervasive 222,386-CpG raw-DM list to 12,922 silencing pairs (4,483 unique genes), recovers ccRCC literature TSGs at 2.55× hypergeometric enrichment ($p = 1.4 \times 10^{-4}$; modest in magnitude, see §4.2), and identifies VHL/`cg13672843` as a clear subgroup-driven silencing event (32 of 318 tumors, ~33% lower VHL mRNA, $p = 4.5 \times 10^{-6}$). A complementary descriptive observation: ~82% of silencing CpGs are patient-subgroup-specific ($\hat{\pi}_{\text{resp}} < 0.3$) rather than cohort-wide, suggesting that cancer methylation studies should routinely report per-tumor responder fractions rather than rely on cohort means alone. The entire approach is built from in-scope course primitives (paired moderated t, partial-correlation regression, BH FDR, sign-flipping permutation null, GSEA/ORA); the contribution is *composition* of established statistical building blocks rather than novel inferential machinery.

Code and data availability

TCGA-KIRC and TCGA-KIRP DNA methylation 450K β-values, RNA-seq STAR-Counts, and clinical metadata were retrieved through `TCGAbiolinks` (Bioconductor 3.22). All analysis scripts, intermediate outputs, random seeds, and `sessionInfo()` records are archived in the project repository.

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